Answer **all** the questions in the spaces provided.

1	**(a)**	State *Newton's second law of motion*.	
			[2]
		The **rate of** change of momentum of a body is (directly) **proportional** to the **net** force acting on it, and, the direction of momentum **change** is in the direction of the **net** force.	
	(b)	Use Newton's second law of motion to deduce the principle of conservation of linear momentum.	
		…………………………………………………………………………………… …………………………………………………………………………………… …………………………………………………………………………………… …………………………………………………………………………………… ……………………………………………………………………………………	[2]
		From Newton's 2nd law of motion, if a system experiences **no net** (external) **force**, (the rate of change of momentum of the system must also be zero). Thus there is no change in **total** momentum of the system / the **total** momentum of a system remains constant / is conserved.	